1 Mark for Review

A triathlon is a multisport race consisting of three different consecutive events. A triathlon participant plans to complete the cycling event with an average speed of 33,264 yards per hour. What is this average speed, in miles per hour? (1 mile = 1,760 yards)

Answer Preview:

2 Mark for Review

An arborist measured and recorded the heights of red maple trees in a certain area. Of the red maple tree heights recorded in this area, the minimum was 46 feet and the maximum was 57 feet. Which inequality is true for all possible values of $t$, where $t$ represents the recorded height, in feet, of a red maple tree in this area?

A $t \leq 46$

B $t \geq 57$

C $t \geq 103$

D $46 \leq t \leq 57$

3 Mark for Review

A capillary tube is used to hold a substance while measuring its melting point. The volume of the substance in this capillary tube is 2.23 cubic millimeters. What is this volume in cubic centimeters? ( 1 centimeter =10 millimeters)

(A) 0.00223

(B) 0.0223

(C) 22.3

(D) 2,230

4 Mark for Review

The measures of the interior angles of triangle $ABC$ are 18°, 36°, and $x$° What is the value of $x$?

(A) 18

(B) 54

(C) 126

(D) 180

5 Mark for Review

What value of $p$ is the solution to the equation $10p - 11p = 11$ ?

(A) $-11$

(B) $-\frac{1}{11}$

(C) $\frac{1}{11}$

(D) $11$

The length of a side of square $X$ is 8 centimeters. The area of rectangle $Y$ is 14 square centimeters. What is the total area, in square centimeters, of square $X$ and rectangle $Y$?

(A) 44

(B) 64

(C) 78

(D) 92

7 Mark for Review

ABC

The distance $d$, in feet, from the ground to a flag on a flagpole $t$ seconds after a person begins lowering it is given by the function $d(t) = 26 - 4t$. In this context, what does 26 represent?

(A) The distance, in feet, of the flag from the ground before it is lowered

(B) The distance, in feet, of the flag from the ground 4 seconds after it begins lowering

(C) The time, in seconds, it takes for the flag to be fully lowered

(D) The rate, in feet per second, at which the flag moves as it is lowered

8 Mark for Review

ABC

The function $f$ is defined by $f(x) = \frac{4}{5}x - \frac{7}{5}$. For what value of $x$ does $f(x) = 5$?

(A) $\frac{9}{2}$

(B) $\frac{32}{5}$

(C) 8

(D) 32

9 Mark for Review

ABC

Which expression is equivalent to $x^{20}\left(x^{4}\right)$ ?

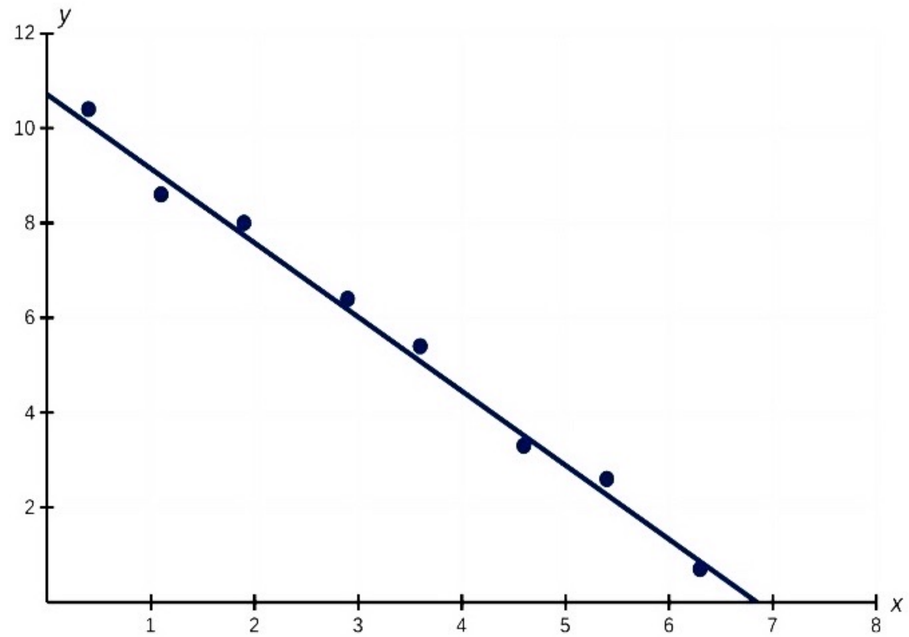
(A) $x^{5}$

(B) $x^{16}$

(C) $x^{24}$

(D) $x^{80}$

In the scatterplot, a line of best fit for the data is shown.




Which of the following equations best represents this line of best fit?

(A) $y = 1.56x$

(B) $y = 10.71 - 1.56x$

(C) $y = 10.71$

(D) $y = -10.71x$

11 Mark for Review

$$8(x-6)^2 = 800$$

What is the positive solution to the given equation?

Answer Preview:

$$x^2 + 27x + 1 = 0$$

What is one of the solutions to the given equation?

(A) $\left[ -27 + \sqrt{\left((27)^2 - 4(27)(27)\right)} \right] / [2(27)]$

(B) $\left[ -27 + \sqrt{\left((27)^2 - 4(1)(1)\right)} \right] / [2(1)]$

(C) $\left[ -1 + \sqrt{\left((1)^2 - 4(1)(27)\right)} \right] / [2(1)]$

(D) $\left[ -1 + \sqrt{\left((1)^2 - 4(27)(1)\right)} \right] / [2(27)]$

13 Mark for Review

The function $g$ is defined by $g(x) = -x / 3 + 11$. What is the $x$-intercept of the graph of $y = g(x) + 5$ in the $xy$-plane?

(A) $(48, 0)$

(B) $(33, 0)$

(C) $(-48, 0)$

(D) $(-33, 0)$

14 Mark for Review

Triangles *FGH* and *JKL* are congruent. In triangle *FGH, FG=2GH, FG=28*, and *FH=29*. In triangle *JKL, JK =2KL*. What is the length of segment *JL* ?

Answer Preview:

15 Mark for Review

$$24x + 7y = 202$$
$$8x - 9y = -182$$

The solution to the given system of equations is $(x, y)$. What is the value of $y$?

Answer Preview:

$$x^2 + y^2 - 8x + 4y - 29 = 0$$

In the $xy$-plane, the graph of the given equation is a circle. What are the coordinates $(x, y)$ of the center of the circle?

(A) $(-8, -29)$

(B) $(-8, 4)$

(C) $(4, -2)$

(D) $(4, 2)$

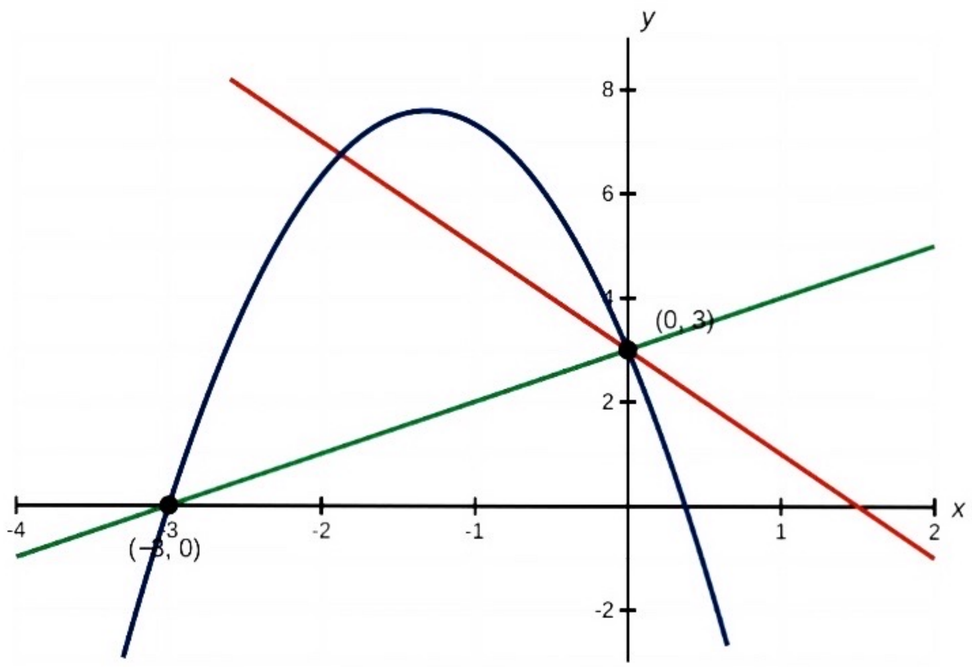
The elastic potential energy E, in newton-meters, stored in a spring is the product of $\frac{1}{2}$ the spring constant, in newtons per meter, and the square of the number of meters, x, the spring is stretched from its equilibrium position. If the spring constant for a certain spring is 74 newtons per meter, which equation gives the elastic potential energy, in newton-meters, stored in this spring in terms of the number of meters the spring is stretched from its equilibrium position?

(A) $E = 148x$

(B) $E = 148x^2$

(C) $E = 37x$

(D) $E = 37x^2$




The graph of a system of three equations is shown. The system consists of two linear equations and one quadratic equation. How many solutions does this system have?

A 1

B 2

C 3

D 4

A scientist initially measures 9,000 bacteria in a growth medium. 8 hours later, the scientist measures 18,000 bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria $t$ hours after the initial measurement. What is the value of r?

(A) $\frac{1}{9,000}$

(B) $\frac{1}{8}$

(C) 8

(D) 9,000

20 Mark for Review

In the xy-plane, lines p and s are perpendicular and intersect at the point $(-3, 4)$. Line p passes through the point $(2, -6)$. . What is the slope of line s?

Answer Preview:

0.0048 is 0.8% of **a**, and 84 is 350% of **b**, where **a** and **b** are positive numbers. What percentage of **a** is **b** ?

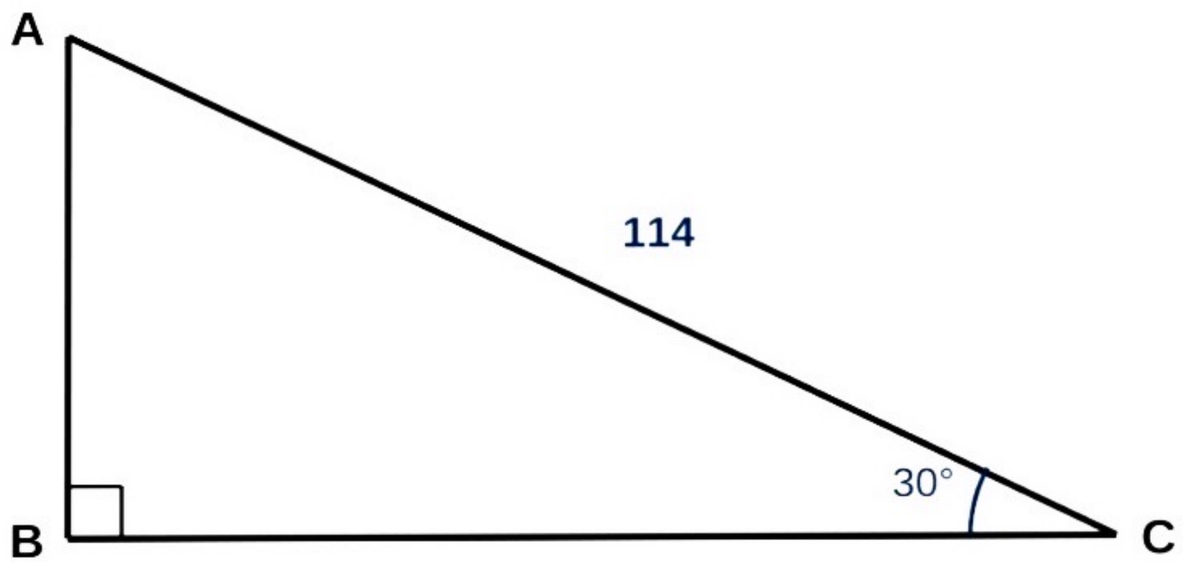
(A) 40%

(B) 144%

(C) 1,440%

(D) 4,000%

22 Mark for Review




Note: figure not drawn to scale.

In the right triangle shown, what is the value of $\tan A$ ?

Answer Preview:

1 Mark for Review

A line passes through the points $(3, 7)$ and $(5, 14)$ in the $xy$-plane. What is the slope of the line?

Answer Preview:

The function $A$ defined by $A(x) = x(x + 2)$ gives the area, in square inches, for a parallelogram with a height of $x$ inches and a base that is 2 inches longer than its height. Which of the following is the best interpretation of $A(50) = 2,600$ in this context?

(A) The area of this parallelogram is 50 square inches when its height is 2,600 inches.

(B) The area of this parallelogram is 52 square inches when its height is 2,600 inches.

(C) The area of this parallelogram is 2,600 square inches when its height is 50 inches.

(D) The area of this parallelogram is 2,600 square inches when its height is 52 inches.

| x | f(x) |
|---|---|
| -1 | 5 |
| 0 | 15 |
| 1 | 27 |

For the quadratic function $f$, the table shows three values of $x$ and their corresponding values of $f(x)$. Which equation defines $f$?

A $f(x) = 11x^2 + x + 15$

B $f(x) = 6x^2 + 6x + 15$

C $f(x) = 9x^2 - x + 15$

D $f(x) = x^2 + 11x + 15$

4 Mark for Review

If $6(9-2x)+3=5(9-2x)+16$, what is the value of $9-2x$ ?

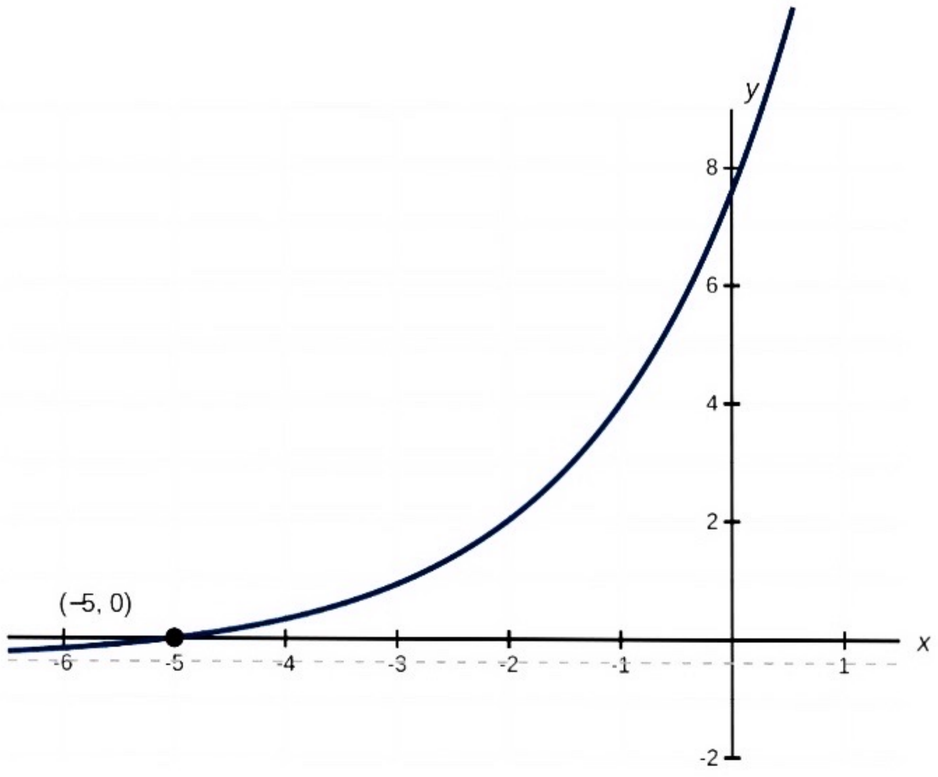
(A) $-13$

(B) $-2$

(C) 2

(D) 13

5 Mark for Review



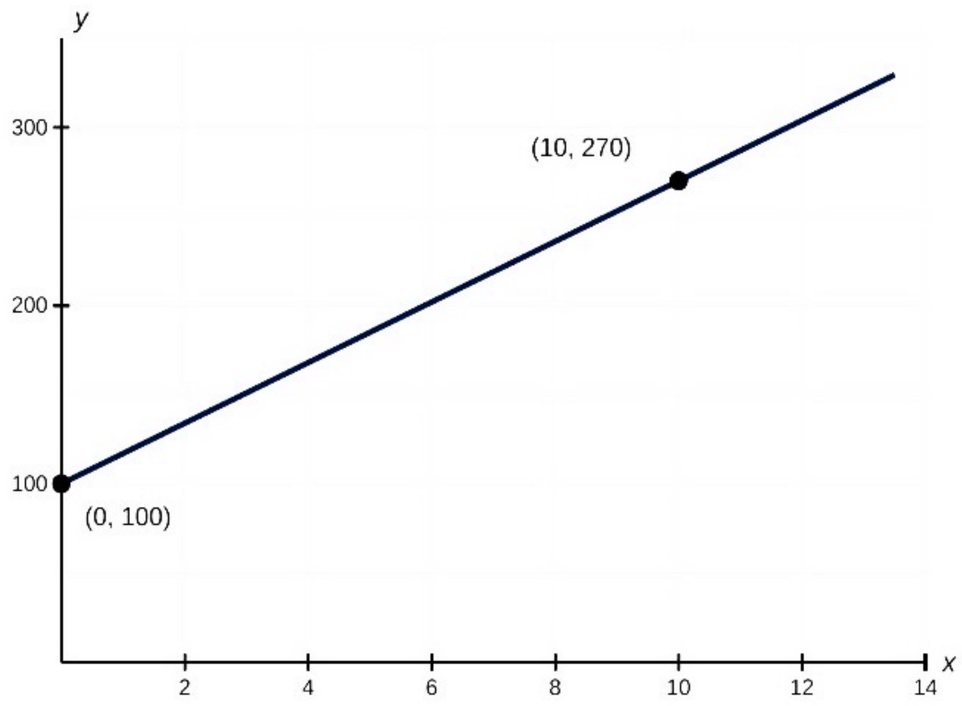

What is the $x$-intercept of the graph shown?

A $(-6, 0)$

B $(-5, 0)$

C $(-4, 0)$

D $(7.6, 0)$




$x$= number of games, $y$= total cost (dollars)

The graph of the function $f$, where $y = f(x)$, gives the total cost $y$, in dollars, for a certain video game system and $x$ games. What is the best interpretation of the slope of the graph in this context?

A Each game costs $17.

B The video game system costs $17.

C Each game costs $100.

D The video game system costs $100.

For the first 10 seconds of an experiment that lasts 100 seconds, a particle travels at a speed that is at least 4 meters per second and is at most 8 meters per second. For the remainder of the experiment, the particle travels at a constant speed of 12 meters per second. Which of the following is a possible number of meters the particle traveled in the first 70 seconds of the experiment?

(A) 725

(B) 766

(C) 801

(D) 840

8 Mark for Review

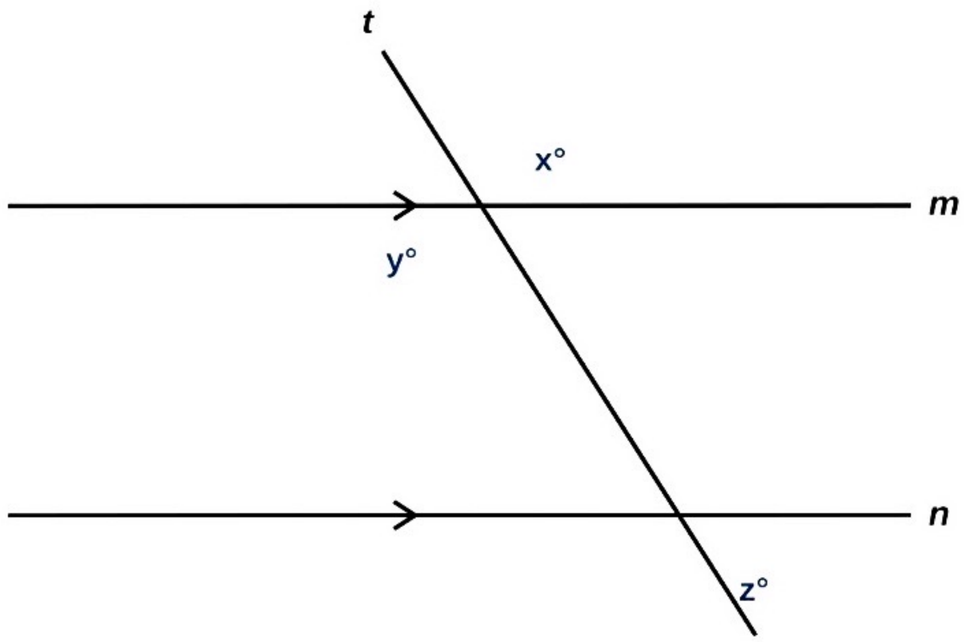
The combined original price for a towel and chair is \$40. After a 45% discount to the towel and a 25% discount to the chair are applied, the combined sale price for the two items is \$28. Which system of equations gives the original price $t$, in dollars, of the towel and the original price $c$, in dollars, of the chair?

A $t + c = 40; 0.45t + 0.25c = 28$

B $t + c = 40; 0.55t + 0.75c = 28$

C $t + c = 40; 0.25t + 0.45c = 28$

D $t + c = 40; 0.75t + 0.55c = 28$




Note: figure not drawn to scale.

In the figure, line $m$ is parallel to line $n$, line $t$ intersects both lines, and $180 < x + y + z < 242$. Which of the following could NOT be a value of $x$?I. 6 II. 61 III. 62

A II only

B III only

C A I and II only

D II and III only

$$2a = \left(4b^2 / a\right) - 47a$$

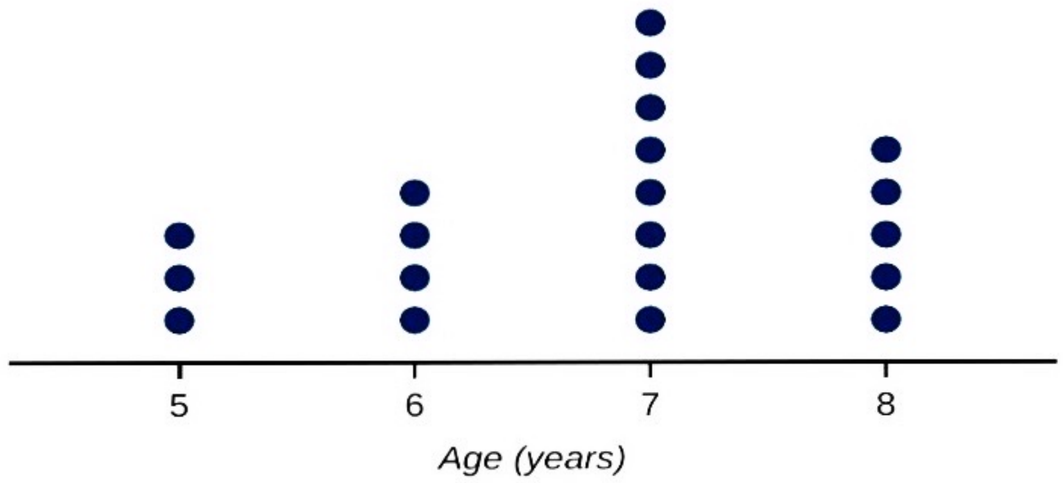
The given equation relates the positive variables $a$ and $b$. Which equation correctly expresses $a$ in terms of $b$?

A $a = 2b / 7$

B $a = 2b^2 / 7$

C $a = 4b / 49$

D $a = 4b^2 / 49$




The dot plot represents an original data set of the ages of the 20 cards in a baseball card collection. A card that is 12 years old is added to the collection, resulting in a new data set of the ages of 21 cards. Which statement best compares the median of the original data set and the median of the new data set?

A The median of the original data set is less than the median of the new data set.

B The median of the original data set is equal to the median of the new data set.

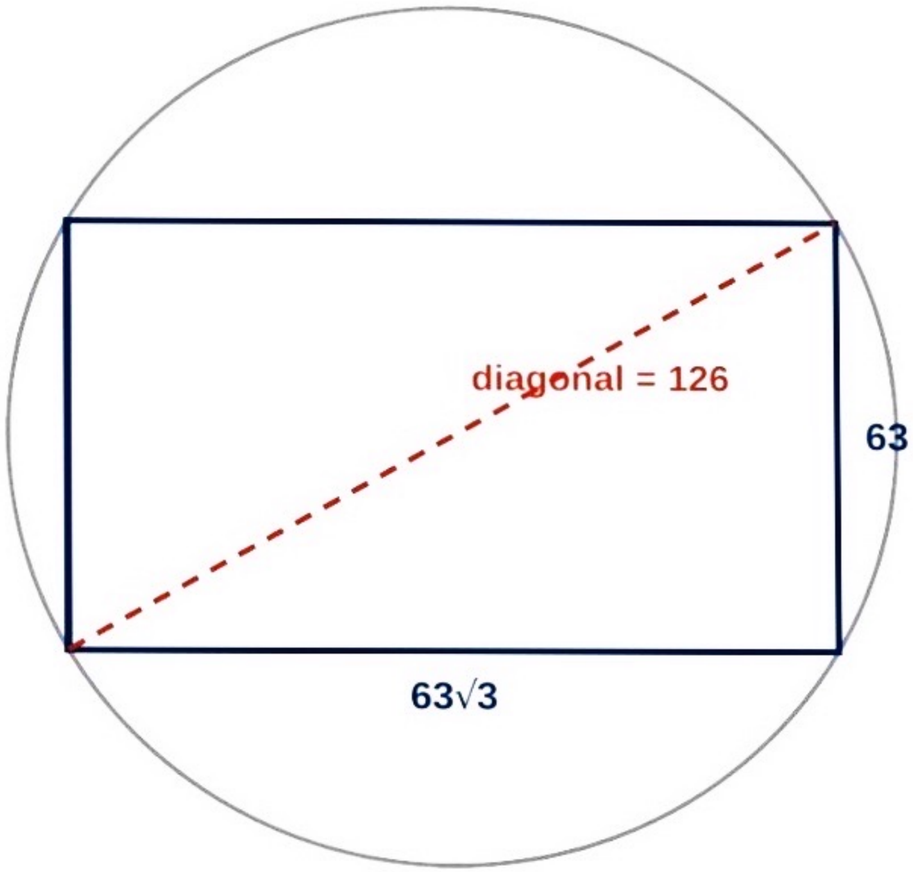
C The median of the original data set is greater than the median of the new data set.

D There is not enough information to compare the medians of the data sets.

12 Mark for Review

The floor of a store has an area of 4,400 square feet. An architect creates a scale model of the floor of the store, where the length of each side of the model is $\frac{1}{20}$ times the length of the corresponding side of the floor of the actual store. What is the area, in square feet, of the scale model?

Answer Preview:




A rectangle is inscribed in a circle, such that the diagonal of the rectangle is twice the length of its shortest side. The area of the circle is $3,969\pi$ square units. What is the perimeter, in units, of the rectangle?

(A) $126 + 126\sqrt{2}$

(B) $126 + 126\sqrt{3}$

(C) 378

(D) 3,969

14 Mark for Review

$$(x + a) / \left(x^2 - a^2\right) = -2 / (25x)$$

In the given equation, $a$ is a constant. The solution to this equation is 6 . What is the value of $a$ ?

Answer Preview:

$$\left(x^{11}y^{5}\right) / \left[5x^{2} \cdot \sqrt[8]{\left(y^{8}\right)}\right]$$

For all positive values of $x$ and $y$, the given expression is equivalent to which of the following?

I. $\left(x^{8/3}\right)\left(y^{5/3}\right) / (5xy)$
II. $\left(x^{5}y^{2}\right) / 5$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

As a car approaches an intersection, a linear function $f$ models its speed, in miles per hour (mph), $x$ seconds after the brake is initially applied. According to the function, the car's estimated speed is 23 mph when the brake is initially applied, and the car's estimated speed decreases by 4.1 mph every second after the brake is initially applied. Which equation defines $f$?

A $f(x) = 23x$

B $f(x) = 4.1x$

C $f(x) = -4.1x + 23$

D $f(x) = -23x + 4.1$

A researcher surveyed a random sample of 1,380 people from a certain state to estimate the percentage of people who support the governor's recent approval of a bill. From the survey, the researcher estimated that 72% of people from the state support the governor's approval of the bill, with an associated margin of error of 2.37%. The researcher repeated the survey with a random sample of people from the state that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

(A) The margin of error is between 2.37% and 3.37%.

(B) The margin of error is between 4.37% and 5.37%.

(C) The margin of error is greater than 5.37%.

(D) The margin of error is less than 2.37%.

A set designer is hammering nails along the top edge of a wall with a length of $6x$ feet, where $x$ is an integer, that will be used to hang garland. The designer hammers the first nail at the left edge of the wall. The designer then hammers an additional nail every $\frac{6}{11}$ feet along the entire length of the wall, starting from the first nail and with the last nail being hammered at the right edge of the wall. Which equation best represents this situation, where $y$ is the number of nails the designer hammers along the entire length of the wall?

(A) $y = 11x + 1$

(B) $y = 11x$

(C) $y = \frac{6}{11}x + 1$

(D) $y = \frac{6}{11}x$

19 Mark for Review

$$8(x-k)^2-2t+33=0$$

In the given equation, $k$ and $t$ are integer constants. The equation has no real solution. What is the greatest possible value of $t$?

Answer Preview:

The function $f$ is defined by $f(x) = a\left(2.1^x + 2.1^b\right)$, where $a$ and $b$ are integer constants and $0 < a < b$. The functions $g$ and $h$ are equivalent to function $f$, where $k$ and m are constants. Which of the following equations displays the $y$-coordinate of the $y$-intercept of the graph of $y = f(x)$ in the $xy$-plane as a constant or coefficient?

I. $g(x) = a\left(2.1^x + k\right)$

II. $h(x) = a(2.1)^x + m$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

21 Mark for Review

The function $h$ is defined by $h(x) = -\sqrt{(x^2 + bx + c)}$, where $b$ and $c$ are constants. In the $xy$-plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{314})$. If $h(m) = 0$, what is the greatest possible value of $m$?

Answer Preview:

In 2005, Chloe earned 12% more than in 2004, and in 2006 Chloe earned 5% more than in 2005. If Chloe earned $y$ times as much in 2004 as in 2006, which of the following is closest to the value of $y$?

(A) 0.4167

(B) 0.6000

(C) 0.8503

(D) 1.1760